Axles and alignment

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements and observations to improve straight rolling.

SUCCESS CRITERIA - I can...

1. Check axle parallelism and wheel clearance.
2. Record a repeatable alignment test.
3. Use evidence to choose one adjustment.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 50-turn coil has flux per turn change by 0.007 Wb in 0.5 s. Find average induced-emf magnitude.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: clearance = available gap - part width.
Use positive clearance and check free rotation. Alignment and wheel rubbing are separate checks; change one feature at a time.

Worked reasoning: Parallel axles can still cause drag if wheels rub the chassis. Alignment and clearance are separate checks; a straight top view does not verify free rotation.

Your turn: Draw a top view and a side detail marking an axle direction and a wheel gap.



Marzano check - circle: ① ② ③

03 Read and interpret the model

On your top-view sketch, mark both axle lines. Add a check for parallelism and show clearance between each wheel and the chassis.



Marzano check - circle: ① ② ③

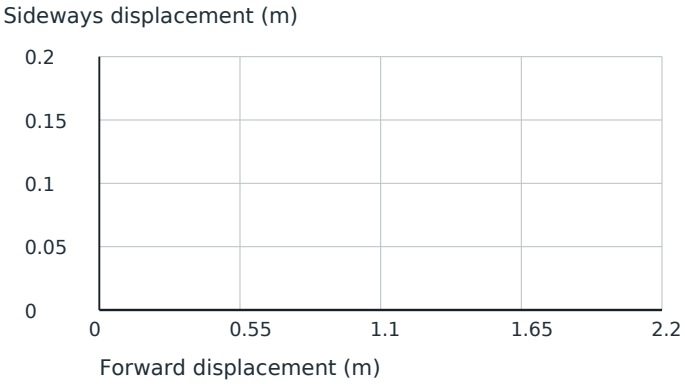
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Continue down each column.

04 Apply the physics

A sample straight-tracking test ends 2.0 m forward and 0.12 m right of the starting point. Plot the endpoint. Calculate the displacement magnitude and direction using $\tan^{-1}(0.12/2.0)$. Explain what these measurements say about straight tracking.



Marzano check - circle: 1 2 3

05 Explain your reasoning

A car veers right once. Does that prove the rear axle is crooked? Name another possible cause and a discriminating check.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 Ω resistor carries 6 A. Find power using both IV and I^2R, showing the implied voltage.
Label input, useful output and other energy transfers.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Two designs complete three identical free-roll tests. Before adjustment, offsets are 0.18, 0.22 and 0.20 m right. After adjustment, they are 0.04, 0.06 and 0.05 m right. Compare mean offset and range. Explain why this is stronger evidence than one excellent run.

Trial	Before offset (m)	After offset (m)
1	0.18	0.04
2	0.22	0.06
3	0.2	0.05

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Repeated 3 m runs finish 0.20, 0.24, 0.22 m left of the line. Find mean deviation and range; design a controlled adjustment and a fair comparison after it.



Marzano check - circle: 1 2 3